YOUSSEF EL BAJOURY

Full Stack Software Engineer | IoT | AI | Cloud

Casablanca, Morocco | +212 688-039822 | youssefelbajoury321@gmail.com
linkedin.com/in/youssef-el-bajoury-192451375 | github.com/youssefelbajoury

SUMMARY

Full Stack Software Engineer IoT, AI, Cloud, with hands-on experience building connected systems end to end — from ESP32 firmware and sensor integration to REST APIs, real-time WebSocket communication and TypeScript dashboards. Sole developer of a complete IoT health-monitoring platform spanning firmware, backend, web dashboard and Flutter mobile client. Available immediately for a permanent position or pre-hire internship in Casablanca.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Dart, PHP, C/C++ (Arduino)
Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3, responsive design
Backend: Node.js, Express.js, REST API design, JWT authentication, WebSocket, Laravel
Mobile: Flutter, Dart
Databases: MongoDB, MySQL
IoT & Embedded: ESP32, Arduino, Raspberry Pi, RFID RC522, sensor integration
Tools: Git, GitHub, Postman
Methodologies: Agile Scrum, UML, MERISE, modular layered architecture

EXPERIENCE

Software Engineer — Smart Health Room (Final Year Project) **Casablanca, Morocco**
Faculté des Sciences Aïn Chock (FSAC) *Sep 2025 – Jul 2026*

- Designed and built, as sole developer, an end-to-end IoT monitoring platform spanning ESP32 firmware, a Node.js/Express REST API, a MongoDB data layer, a Next.js/TypeScript web dashboard and a Flutter mobile client consuming the same API.
- Integrated 6 sensors (temperature, humidity, gas, luminosity, pressure, presence) and streamed readings from ESP32 to the backend over WebSocket, updating the dashboard in real time without page reload.
- Implemented JWT authentication with two-tier role-based access control (administrator, user), plus RFID RC522 badge authentication for physical room access.
- Delivered role-specific interfaces covering live telemetry, historical data visualization, device control and user management, mirrored across web and mobile.
- Structured the codebase as a modular layered architecture (firmware / API / data / presentation) across 8 Git repositories, allowing new sensors and interfaces to be added without modifying existing modules.
- Tested and validated REST endpoints with Postman throughout development.

Full Stack Developer — Internship **Soualem, Morocco**
CFIT *Feb 2025 – Mar 2025*

- Contributed full stack to a restaurant reservation web application within a 5-developer Agile team, working across the React.js interface and the Node.js/Express backend.
- Developed REST API endpoints backed by MongoDB and integrated them into the React frontend.
- Collaborated through a branch-based Git workflow and Agile ceremonies over a full-time internship.

EDUCATION

Licence Professionnelle — Advanced Development for IoT and Intelligent Systems (DAISI) **Casablanca**
Faculté des Sciences Aïn Chock (FSAC) *Sep 2025 – Jul 2026*
Coursework: Full Stack development, IoT and embedded systems, mobile development (Flutter), software architecture, UML/MERISE modeling, Agile Scrum.

Diplôme de Technicien Spécialisé — Digital Development, Full Stack Option **Casablanca**
OFPPT *Sep 2023 – Jun 2025*
Coursework: JavaScript, React.js, Node.js, Express.js, PHP, Laravel, MongoDB, MySQL, REST API design, UML modeling, Git.

Baccalauréat — Physical Sciences, mention Bien **Casablanca**
Farabi High School *2023*

LANGUAGES

Arabic — Native • French — Fluent (medium of instruction throughout higher education) • English — Professional working proficiency (Altissia B1)